Branching Logic

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Branching Logic before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Decompose a Goal

Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Practise with fading help

> Complete the supported Branching Logic trace and record the first step you can explain without help.

Use W2: Trace the tested path for **Branch**ing and **boundary** tests. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Branching Logic explanation against the supplied facts before discussing it.

Use the worked case for **Branch**ing and **boundary** tests. Proposed explanation: “Every **condition** in the chain always runs”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Build a new sorter independently

Use an integer tiles count. Label a negative value Invalid, zero Empty, 1 through 6 Small and values above 6 Large. Write an **order**ed if elif else script and predict -1, 0, 1, 6 and 7. Include one print after the chain.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Branching Logic target to a trace and a justified conclusion.

> Classify lengths with if pages<80 then Short; else if pages<=160 then Medium; else Long. Trace 79,80,160,161. What happens if the <=160 **branch** is placed first?

Solve the given case

> Solve the supplied Branching Logic case and show the deciding evidence.

If pages<100 choose Short; otherwise if pages<=200 choose Medium; otherwise Long.

Classify 99,100,200,201 pages.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

If pages are at most 80, choose short; otherwise choose long. Classify books with 79, 80 and 81 pages and explain the **boundary**.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 3 Test and reflect

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, **boundary** case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	

Test record	My evidence
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Branching Logic adds to Transparent Book Recommender and keep the evidence.

Branch when a reader has a hard constraint such as language, format, or maximum length.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.

- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Classify lengths with if pages<80 then Short; else if pages<=160 then Medium; else Long. Trace 79,80,160,161. What happens if the <=160 **branch** is placed first?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the chain

> Explain the chain. Apply the supplied Branching Logic facts and keep your reasoning.

Describe the purpose of if, elif and else in the worked duration sorter. Explain why print(label) is outside the indented **branch**es.

> **My reasoning and evidence**

W2 Trace the tested path

> Trace the tested path. Apply the supplied Branching Logic facts and keep your reasoning.

For -1, 12 and 13, record checked **condition**s, skipped **condition**s and the final label. Run each as a fresh script after predicting.

> **My reasoning and evidence**

W3 Repair the overwritten result

> Repair the overwritten result. Apply the supplied Branching Logic facts and keep your reasoning.

Trace the two-independent-if example for -1. Identify which assignment overwrites the earlier label and make the smallest repair.

My reasoning and evidence

W4 Build a new sorter independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Build a new sorter independently. Apply the supplied Branching Logic facts and keep your reasoning.

Use an integer tiles count. Label a negative value Invalid, zero Empty, 1 through 6 Small and values above 6 Large. Write an **order**ed if elif else script and predict -1, 0, 1, 6 and 7. Include one print after the chain.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Choose revealing regression tests

> Choose revealing regression tests. Apply the supplied Branching Logic facts and keep your reasoning.

A repair changes the small-count **condition** from < 6 to <= 6. Name the case that exposes the original error and two different kinds of earlier cases to rerun.

> **My reasoning and evidence**

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

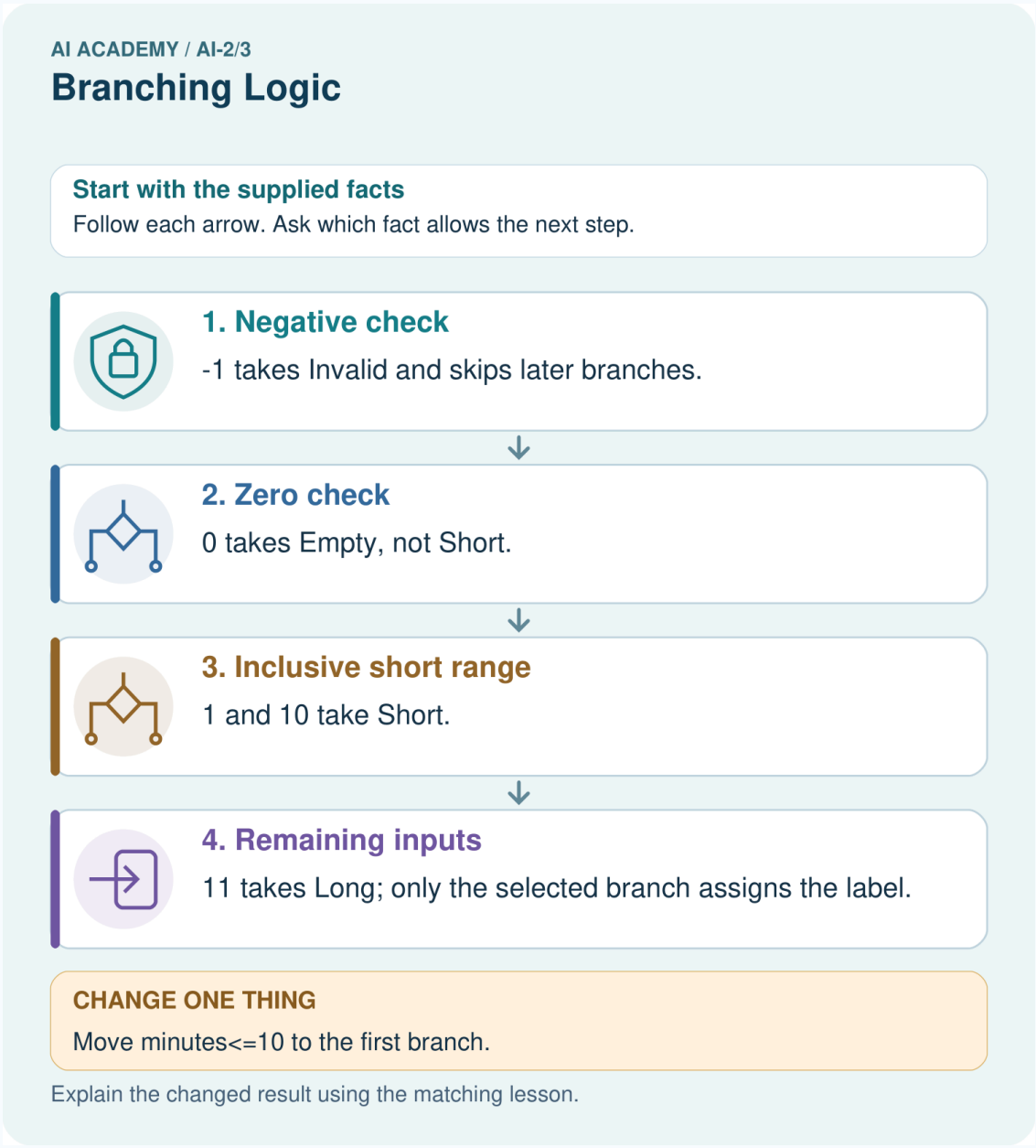
Pictorial case practice: Branching Logic

> A fictional reading-time labeler uses one Python if/elif/else chain. Explain the labelled step before checking the changed condition.

A fictional reading-time labeler uses one Python if/el**if/else** chain.

Whole minutes below 0: Invalid; 0: Empty; 1–10: Short; above 10: Long.

Order checks as minutes<0, minutes==0, minutes<=10, otherwise. Print the assigned label once.



Follow the labelled picture with the matching explanation. Open the vector diagram

Give expected labels for –1,0,10,11 and identify the equality **boundary** for Short.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Validate recommender requests before ranking items, retaining explicit labels for missing or invalid cases.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Validate recommender requests before ranking items, retaining explicit labels for missing or invalid cases.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional reading-time labeler uses one Python if/el**if/else** chain.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? 11 takes Long; only the selected **branch** assigns the label.

1. Negative check

2. Zero check

3. Inclusive short range

4. Remaining **input**s

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Remaining **input**s 11 takes Long; only the selected **branch** assigns the label.

The third comparison alone would include negative values and zero, but earlier **branch**es already handle them. An elif chain selects its first true **branch**.

Multiple independent if statements would have different behavior and could overwrite an earlier label. **Boundary** traces show why **order** belongs to the specification.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Move minutes<=10 to the first **branch**. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. –1 and 0 wrongly receive Short.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed **condition** and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: –1 and 0 wrongly receive Short. Both satisfy the earlier comparison before their intended checks are reached.

Repair a missing building block

Review Decompose a Goal (ai-2/2). Goal: Distinguish a function definition from a function call. Short practice: What happens visibly if the caller stores needed but omits print(needed)?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

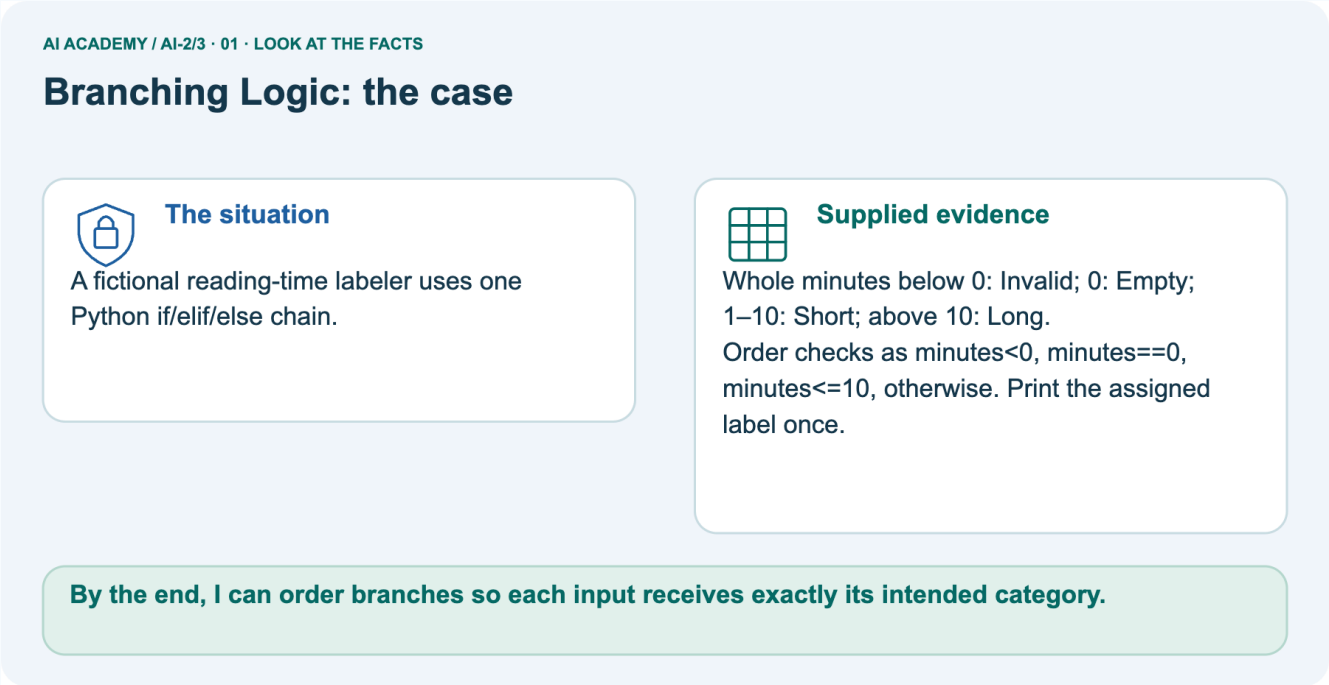
> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

WORKBOOK · 21 OF 21

Visual concept journey

> Explain each Branching Logic picture from its labels, then record a relevant limit.

01. Branching Logic: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

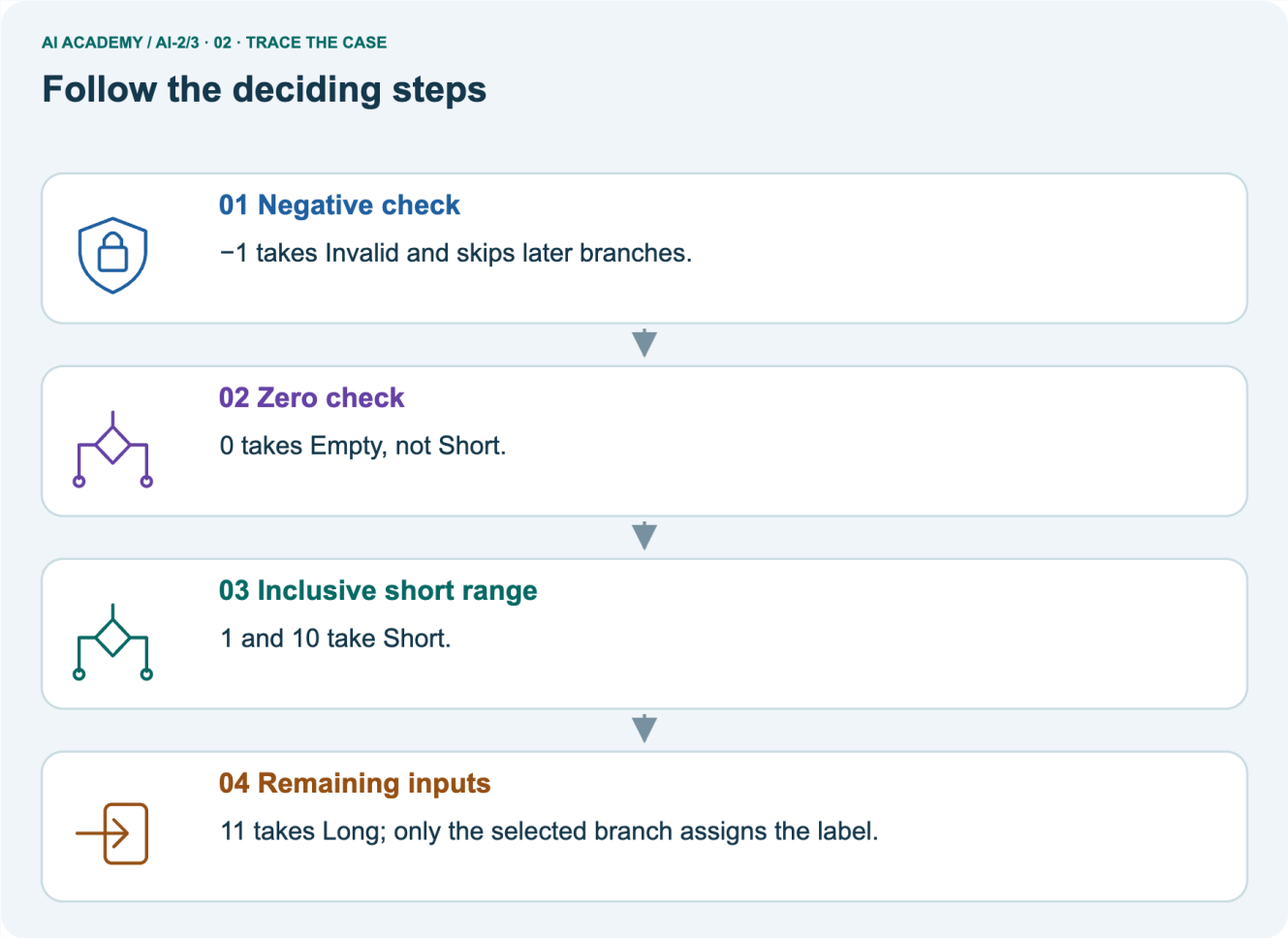
A fictional reading-time labeler uses one Python if/eli**if/else** chain. Supplied facts: Whole minutes below 0: Invalid; 0: Empty; 1–10: Short; above 10: Long. **Order** checks as minutes<0, minutes==0, minutes<=10, otherwise.

Print the assigned label once.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



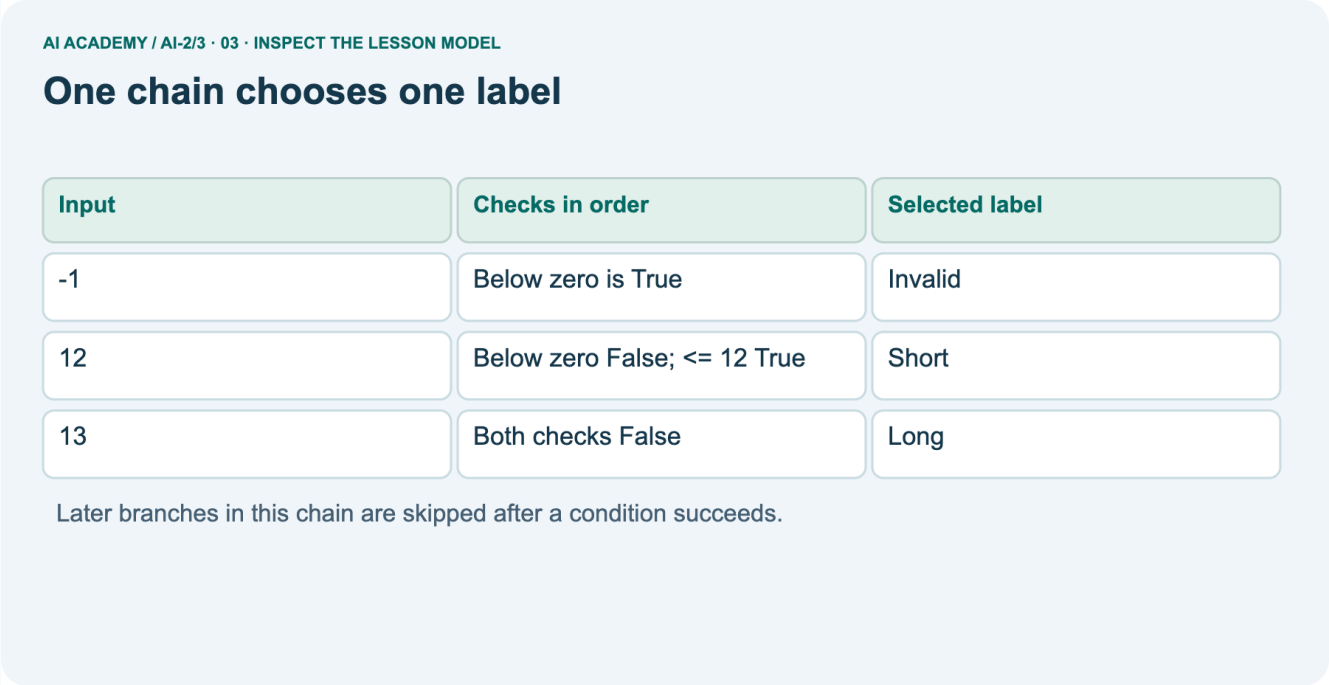
Follow the labelled picture with the matching explanation. Open the vector diagram

Negative check: −1 takes Invalid and skips later **branch**es. → Zero check: 0 takes Empty, not Short. → Inclusive short range: 1 and 10 take Short. → Remaining **input**s: 11 takes Long; only the selected **branch** assigns the label.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. One chain chooses one label



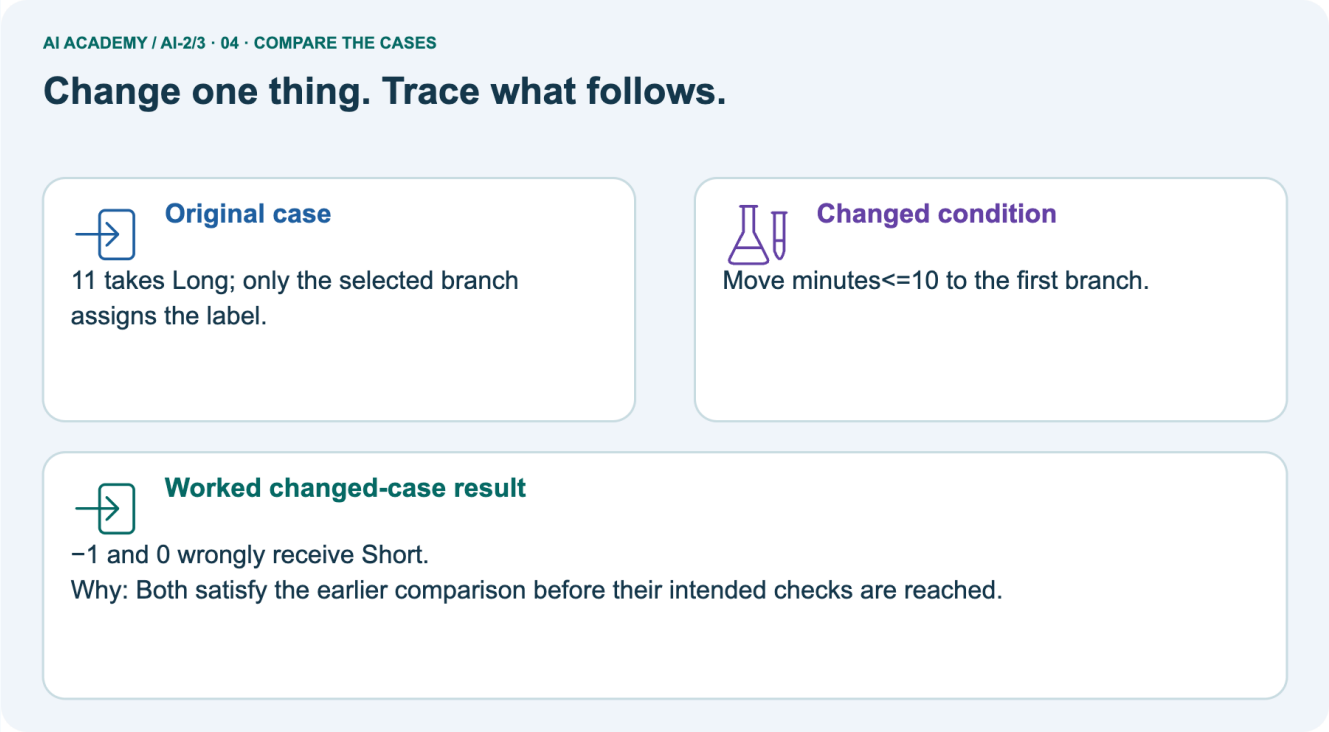
Follow the labelled picture with the matching explanation. Open the vector diagram

One chain chooses one label. -1 | Below zero is True | Invalid. 12 | Below zero False; <= 12 True | Short. 13 | Both checks False | Long

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

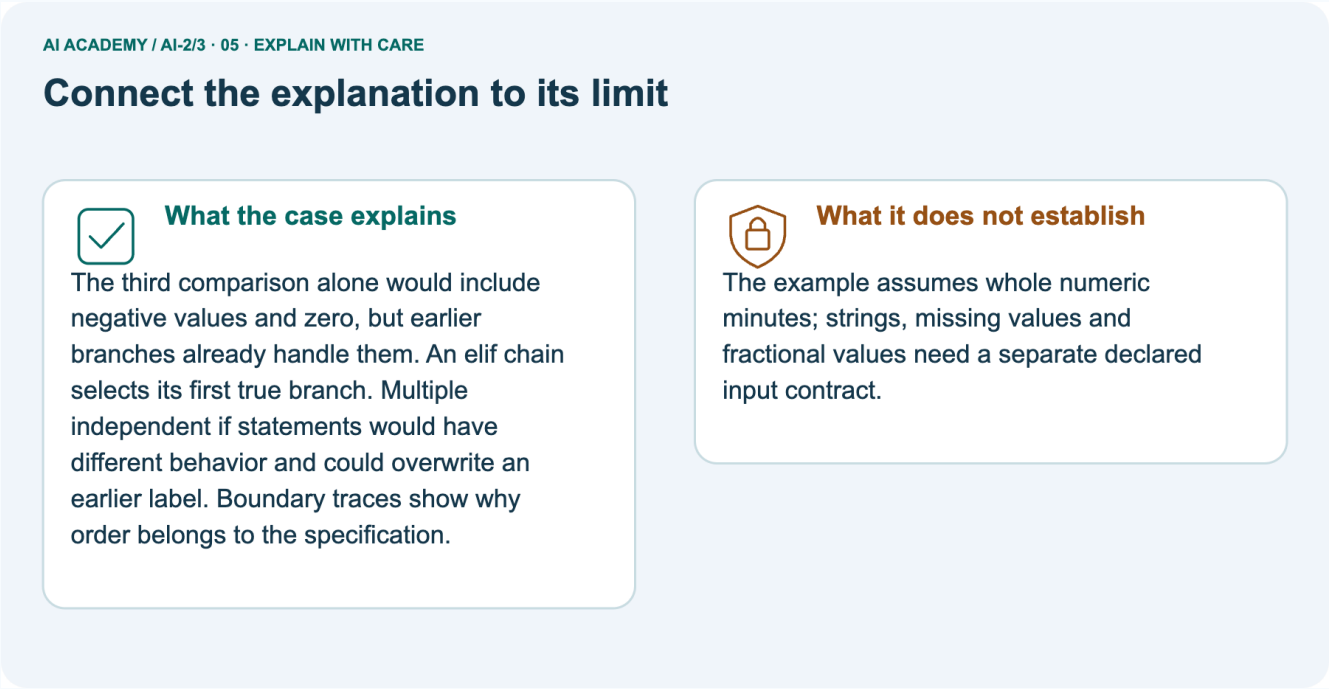
Original case: 11 takes Long; only the selected **branch** assigns the label. Changed **condition**: Move minutes<=10 to the first **branch**. Expected result: −1 and 0 wrongly receive Short.

Reason: Both satisfy the earlier comparison before their intended checks are reached.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The third comparison alone would include negative values and zero, but earlier **branch**es already handle them. An elif chain selects its first true **branch**.

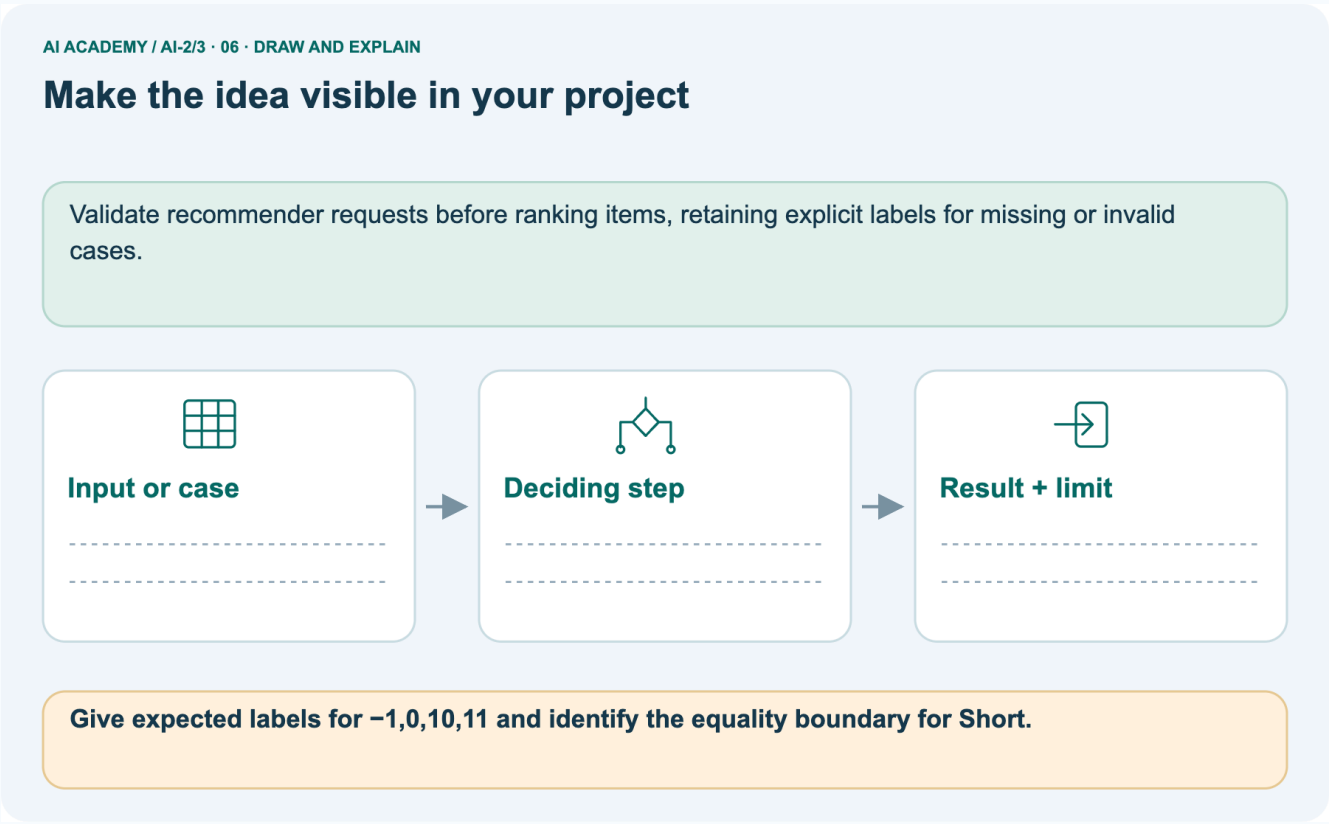
Multiple independent if statements would have different behavior and could overwrite an earlier label. **Boundary** traces show why **order** belongs to the specification.

Limit: The example assumes whole numeric minutes; strings, missing values and fractional values need a separate declared **input** contract.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate recommender requests before ranking items, retaining explicit labels for missing or invalid cases. Draw your **input**, deciding step, result and limit.

Supported question: Give expected labels for −1,0,10,11 and identify the equality **boundary** for Short.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.